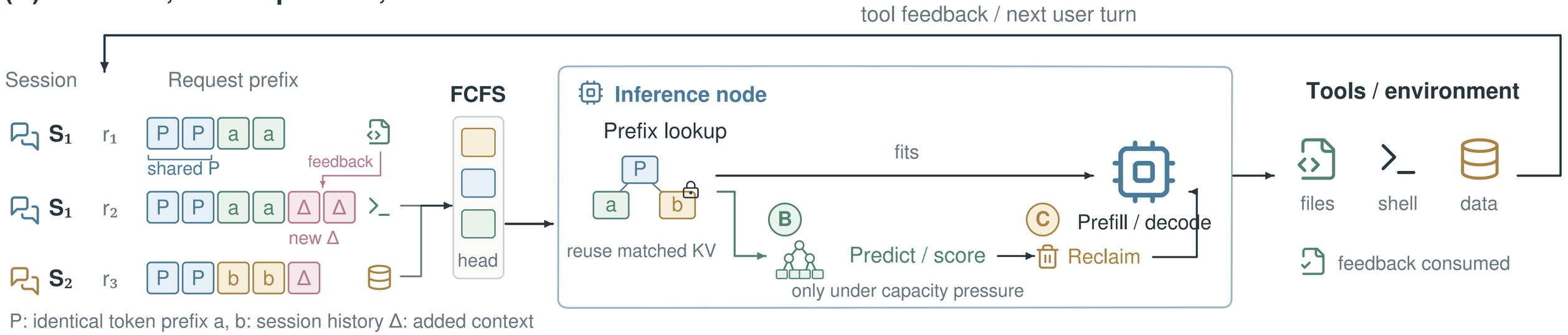
(A) Sessions, shared prefixes, and native inference



P: identical token prefix a, b: session history Δ: added context

(B) Learn a distribution, then score reuse

B

TRAIN · observed snapshots

- Tool identity + phase
- Idle time + tool time
- Queued? + queue length
- Prefix tokens + retries

6 projects · 1,079 snapshots
14 encoded columns, incl. bin k

Risk intervals

state	k	label
x_1	1	wait
x_1	2	wait
x_1	3	U
x_2	…	~~mask~~

9,229 weighted rows

Fit log-loss gradients

Round 1 + Round 2 + … Round 40

$$F_m = F_{m-1} + 0.05\,T_m$$

40 rounds · depth ≤ 2 · ≤ 4 leaves
3 class trees / round: U, D, wait

deploy fit

PREDICT · current state

live x_i → GBDT or empirical → **Hazards** (U, D, wait) → survival recursion → **Demand mass p^U** (H = 30 s; .01, 1, 30, 600; time-bin edge (s)) → **Discount & sum**

$$W_i = \sum_{r_k \le H} e^{-r_k/\tau}\, p_{ik}^{U}$$

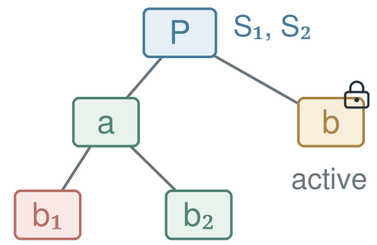
H = 30 s · τ = 10 s
equal cost = 1

C

Optional refit: +100 mature episodes, ≥5 projects → boundary swap

(C) Score → physical KV

$$V_b = \max_{i \in \mathcal{O}_b} W_i / B_b$$



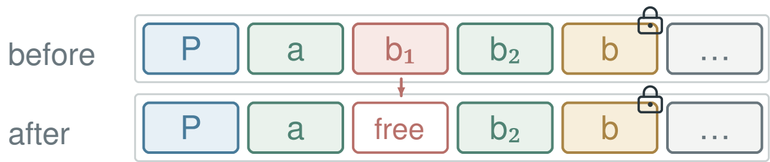
leaf	$V_b B_b$
b_1	0.07
b_2	0.19

lowest legal score

plan: {b_1, …} → rescan until fit

Unbound first · support missing → LRU

Apply completed plan once



Shared P stored once · reserved buffer unchanged